

Signaling Techniques for Green Communications

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Introduction

- Modern communication systems have soaring data rates and huge energy consumption, resulting in increased CO₂ emission and related environmental problems
- To reduce energy consumption and CO₂ emission, and improve energy efficiency, researchers have proposed the “*Green Communications*” concept
- This concept refers to implementation of hardware or software techniques in the ICT sector that aim to reduce energy consumption
- It is motivated by the need to mitigate environmental pollution and reduce economic costs associated with ICT operations

- Key technological aspects of green communications are as follows
- *Cognitive radio*: It can effectively improve spectrum resource utilization efficiency and network transmission performance
- *Network coding*: Redundant routes are removed to improve network throughput and to attain the effect of energy and bandwidth saving
- *Smart grid*: A modern electricity grid with advanced power facilities of measurement, monitoring, protection, and control; an upgraded and improved traditional grid that gives more economy, security, reliability, and environmental protection

- *Device-to-device (D2D) communication*: It is a radio access technology that provides users the ability to communicate directly between them when they are in close proximity, without traversing traffic through the network infrastructure
- *Massive multiple-input multiple-output (MIMO) systems*: Here large numbers of antennas are used at the transmitter and receiver; these have several advantages like good energy efficiency, robustness, enhanced throughput, latency reduction, and high capacity gains
- *Green internet-of-things (IoT)*: To facilitate reduction in greenhouse effect, green IoT plays an important role by employing energy efficient procedures; wireless sensor networks form a key element for IoT

- In green communications, low complexity and low power receivers are needed in a variety of applications, e.g., wireless sensor networks, short range interconnects, cooperative communications, and emerging communication technologies
- Support for IoT has also created new interest in developing low complexity energy efficient communication protocols
- **One way to achieve low complexity and low power consumption is by fully noncoherent (no channel state information [CSI]) or partially noncoherent (partial CSI) communication**

System Model: N -Branch Receive Diversity and L -Level ASK

- Consider a wireless communication system in flat *Rayleigh fading* with N -branch diversity reception, and *one-sided multi-level amplitude-shift keying* (ASK) having L levels, equal a priori probabilities, and symbol-by-symbol detection
- The complex baseband received signal at the k th diversity branch in a symbol interval is given by

$$r_k = h_k s + n_k, \quad k = 1, \dots, N, \quad \text{where}$$

s : information-bearing ASK symbol,

h_k : random complex fading gain of k th diversity branch,

n_k : additive white Gaussian noise (AWGN) at k th diversity branch

- Due to Rayleigh fading, $h_1, \dots, h_N$ are independent and identically distributed (i.i.d.) complex circular Gaussian random variables, each distributed as $\mathcal{CN}(0, \sigma_h^2)$
- $n_1, \dots, n_N$ are i.i.d. complex circular Gaussian random variables, each distributed as $\mathcal{CN}(0, \sigma_n^2)$
- $h_1, \dots, h_N$ are independent of $n_1, \dots, n_N$
- s takes one of L levels from an ASK signal constellation $\mathcal{S} = \{\sqrt{E_1}, \dots, \sqrt{E_L}\}$
- $\sqrt{E_i}$: i th symbol amplitude level,
 E_i : i th symbol energy level, $i = 1, \dots, L$

- The amplitude levels are taken, without loss of generality, to be in ascending order, implying

$$0 \leq \sqrt{E_1} < \sqrt{E_2} < \cdots < \sqrt{E_{L-1}} < \sqrt{E_L}$$

- Equiprobable signaling: $s = \sqrt{E_i}$ with a priori probability $1/L$ for $i = 1, \dots, L$

- *Average symbol energy:* $E_{\text{av}} = \frac{1}{L} \sum_{i=1}^L E_i$

- *Average signal-to-noise ratio (SNR) per symbol per branch:*

$$\Gamma_{\text{av}} = \frac{E_{\text{av}} \sigma_{\text{h}}^2}{\sigma_{\text{n}}^2} = \frac{1}{L} \sum_{i=1}^L \Gamma_i$$

- *Average SNR per branch of the i th symbol:*

$$\Gamma_i = \frac{E_i \sigma_h^2}{\sigma_n^2}, \quad i = 1, \dots, L$$

- $0 \leq \Gamma_1 < \Gamma_2 < \dots < \Gamma_{L-1} < \Gamma_L$
- The objective of the receiver is to detect, from $r_1, \dots, r_N$, which of the L ASK symbols is transmitted
- *Received signal vector:* $\mathbf{r} = [r_1, \dots, r_N]^T$
- *Fading gain vector OR channel vector:*
 $\mathbf{h} = [h_1, \dots, h_N]^T$

- We consider 3 types of receivers:
 - *Fully noncoherent receiver*: The receiver has no knowledge of the N elements $h_1, \dots, h_N$ of the channel vector $\mathbf{h}$ (no CSI); however, it has knowledge of $\mathbf{r}$, $\mathcal{S}$, and the statistical parameters $\sigma_{\mathbf{h}}^2$ and $\sigma_{\mathbf{n}}^2$ of the channel
 - *Channel magnitude based receiver*: The receiver has knowledge of the fading gain magnitudes $|h_1|, \dots, |h_N|$ but not of the fading gain phases $\angle h_1, \dots, \angle h_N$ (partial CSI); furthermore, it has knowledge of $\mathbf{r}$, $\mathcal{S}$, $\sigma_{\mathbf{h}}^2$, and $\sigma_{\mathbf{n}}^2$
 - *Channel norm based receiver*: The receiver has knowledge of $\|\mathbf{h}\|$, but it has no knowledge of the N elements $h_1, \dots, h_N$ of the channel vector $\mathbf{h}$ (partial CSI), and knowledge of $\|\mathbf{r}\|$ and not of the N individual elements of $\mathbf{r}$; furthermore, it has knowledge of $\mathcal{S}$, $\sigma_{\mathbf{h}}^2$, and $\sigma_{\mathbf{n}}^2$

Receiver Structures

- Decision rule for *fully noncoherent optimum receiver* (this is the optimum receiver in the maximum likelihood [ML] sense):

$$\hat{s} = \arg \min_{s \in \mathcal{S}} \left[\frac{1}{(|s|^2 \sigma_h^2 + \sigma_n^2)} \mathbf{r}^H \mathbf{r} + N \ln (|s|^2 \sigma_h^2 + \sigma_n^2) \right]$$

- This can also be interpreted as an energy detector

since $\mathbf{r}^H \mathbf{r} = \|\mathbf{r}\|^2 = \sum_{k=1}^N |r_k|^2$ is proportional to the energy of the received signal

- Decision rule for *channel magnitude based optimum receiver* (this is the optimum receiver in the ML sense):

$$\hat{s} = \arg \max_{s \in \mathcal{S}}$$

$$\left[\sum_{k=1}^N \ln \left\{ I_0 \left(\frac{2|s||h_k||r_k|}{\sigma_n^2} \right) \right\} - \frac{|s|^2}{\sigma_n^2} \sum_{k=1}^N |h_k|^2 \right]$$

- The optimum channel magnitude based receiver has high complexity and has implementation difficulty due to the presence of the Bessel function nonlinearity
- Suboptimum low complexity channel magnitude based receivers are therefore considered

- Decision rule for *weighted envelope detection (WENVD) receiver* (channel magnitude based suboptimum receiver):

$$\begin{aligned}
 \hat{s} &= \arg \max_{s \in \mathcal{S}} \left[\sum_{k=1}^N \frac{2|s||h_k||r_k|}{\sigma_n^2} - \frac{|s|^2}{\sigma_n^2} \sum_{k=1}^N |h_k|^2 \right] \\
 &= \arg \min_{s \in \mathcal{S}} \sum_{k=1}^N \left(|r_k| - |h_k||s| \right)^2 \\
 &= \arg \max_{s \in \mathcal{S}} \left[|s| \sum_{k=1}^N |h_k||r_k| - \frac{|s|^2}{2} \sum_{k=1}^N |h_k|^2 \right]
 \end{aligned}$$

- Decision rule for *energy detection (ED) receiver* (channel magnitude based suboptimum receiver):

$$\begin{aligned}
 \hat{s} &= \arg \min_{s \in \mathcal{S}} \sum_{k=1}^N \frac{\left(|r_k|^2 - |h_k|^2 |s|^2 \right)^2}{|h_k|^2} \\
 &= \arg \min_{s \in \mathcal{S}} \left(\|\mathbf{r}\|^2 - \|\mathbf{h}\|^2 |s|^2 \right)^2 \\
 &= \arg \max_{s \in \mathcal{S}} \left[|s|^2 \sum_{k=1}^N |r_k|^2 - \frac{|s|^4}{2} \sum_{k=1}^N |h_k|^2 \right]
 \end{aligned}$$

- The ED receiver structure is very similar to that of the fully noncoherent optimum receiver, allowing straightforward switching between the two

- Decision rule for optimum *channel norm based energy detection (CNBED)* receiver:

$$\hat{s} = \arg \max_{s \in \mathcal{S}} \left[-\frac{\|\mathbf{h}\|^2 s^2}{\sigma_n^2} + \ln \left\{ \frac{I_{N-1} \left(\frac{2\|\mathbf{r}\| \cdot \|\mathbf{h}\| \cdot s}{\sigma_n^2} \right)}{\left(\frac{\|\mathbf{r}\| \cdot \|\mathbf{h}\| \cdot s}{\sigma_n^2} \right)^{N-1}} \right\} \right]$$

- From the central limit theorem, *the conditional distribution of $\|\mathbf{r}\|^2$, conditioned on $\|\mathbf{h}\|$ and s , can be approximated as*

$$\mathcal{N}(N\sigma_n^2 + |s|^2\|\mathbf{h}\|^2, N\sigma_n^4 + 2\sigma_n^2|s|^2\|\mathbf{h}\|^2)$$

- This Gaussian approximation is good for

$$N \gg 1 \quad \text{or} \quad |s|^2\|\mathbf{h}\|^2 \gg \sigma_n^2 \quad \text{or} \quad \text{both}$$

- Using the Gaussian approximation, the optimum CNBED receiver decision rule is approximated as

$$\hat{s} \approx \arg \max_{s \in \mathcal{S}} \left[\begin{array}{l} -\frac{\left(\|\mathbf{r}\|^2 - \{N\sigma_n^2 + |s|^2 \|\mathbf{h}\|^2\} \right)^2}{2\sigma_n^2 (N\sigma_n^2 + 2|s|^2 \|\mathbf{h}\|^2)} \\ -\ln \left(\sqrt{N\sigma_n^2 + 2|s|^2 \|\mathbf{h}\|^2} \right) \end{array} \right]$$

- Decision rule for *suboptimum CNBED (SCNBED)* receiver (with parameter α , $\alpha \in (-\infty, \infty) \setminus \{0\}$):

$$\begin{aligned}\hat{s} &= \arg \min_{s \in \mathcal{S}} \left(\|\mathbf{r}\|^\alpha - \|\mathbf{h}\|^\alpha |s|^\alpha \right)^2 \\ &= \arg \max_{s \in \mathcal{S}} \left[\|\mathbf{r}\|^\alpha |s|^\alpha - \frac{1}{2} \|\mathbf{h}\|^\alpha |s|^{2\alpha} \right]\end{aligned}$$

- This suboptimum receiver *compares* $\|\mathbf{r}\|^\alpha$ with $\|\mathbf{h}\|^\alpha |s|^\alpha$, which is done by taking the difference and squaring, and the symbol $\hat{s}$ that gives the minimum squared difference is the detector output
- When $\alpha = 2$, we get the ED receiver

Symbol Error Probabilities

- Using a cumulative distribution function approach, a closed form expression for the symbol error probability (SEP) $P_{\text{fnch,e}}$ of the fully noncoherent receiver is obtained as

$$P_{\text{fnch,e}} = \frac{(L-1)}{L} + \frac{1}{L} \sum_{i=1}^{L-1} e^{-\frac{N \ln(C_i)}{(1-1/C_i)}} \sum_{k=0}^{N-1} \frac{\left(\frac{N \ln(C_i)}{(1-1/C_i)} \right)^k}{k!} - \frac{1}{L} \sum_{i=1}^{L-1} e^{-\frac{N \ln(C_i)}{(C_i-1)}} \sum_{k=0}^{N-1} \frac{\left(\frac{N \ln(C_i)}{(C_i-1)} \right)^k}{k!}, \quad \text{where}$$

$$C_i = \frac{(\Gamma_{i+1} + 1)}{(\Gamma_i + 1)}, \quad i = 1, \dots, L-1$$

- Under the high SNR conditions

$$\Gamma_{\text{av}} \gg 1,$$

$$\Gamma_1 = 0, \quad \Gamma_i \gg 1, \quad i = 2, \dots, L,$$

we can approximate the SEP $P_{\text{wenvd,e}}$ of the WENVD receiver in closed form as $P_{\text{wenvd,e}} \approx P_{\text{wenvd,e,hiSNR}}$, where

$$P_{\text{wenvd,e,hiSNR}} = \frac{1}{L} \binom{2N-1}{N-1} \sum_{i=1}^{L-1} \left[\left\{ \frac{4(\sqrt{\Gamma_{i+1}} + \sqrt{\Gamma_i})^2}{[(\sqrt{\Gamma_{i+1}} + \sqrt{\Gamma_i})^2 - 4\Gamma_i]^2} \right\}^N + \left\{ \frac{4(\sqrt{\Gamma_{i+1}} + \sqrt{\Gamma_i})^2}{[4\Gamma_{i+1} - (\sqrt{\Gamma_{i+1}} + \sqrt{\Gamma_i})^2]^2} \right\}^N \right]$$

- Under the high SNR condition

$$(\Gamma_{i+1} - \Gamma_i) \gg 1, \quad i = 1, \dots, L-1,$$

along with the condition

$$\Gamma_i \gg N, \quad i = 1, \dots, L, \quad \text{or}$$

$$\Gamma_1 = 0, \quad \Gamma_i \gg N, \quad i = 2, \dots, L,$$

we can approximate the SEP $P_{\text{ocnbed,e}}$ of the optimum CNBED receiver in closed form as $P_{\text{ocnbed,e}} \approx P_{\text{ocnbed,e,hiSNR},\Gamma_2,\dots,\Gamma_L \gg N}$, where

$$P_{\text{ocnbed,e,hiSNR},\Gamma_2,\dots,\Gamma_L \gg N} = \frac{2}{L} \binom{2N-1}{N-1} \sum_{i=1}^{L-1} \frac{1}{(\sqrt{\Gamma_{i+1}} - \sqrt{\Gamma_i})^{2N}}$$

- Using a characteristic function approach, a *closed form expression for the SEP* $P_{\text{scnbed,e}}$ *of the SCNBED receiver* is obtained as

$$P_{\text{scnbed,e}} = \frac{1}{L} \sum_{i=1}^{L-1} \left[\delta_{i,-}^N \sum_{k=0}^{N-1} \binom{N+k-1}{k} (1 - \delta_{i,-})^k + \delta_{i,+}^N \sum_{k=0}^{N-1} \binom{N+k-1}{k} (1 - \delta_{i,+})^k \right], \quad \text{where}$$

$$\delta_{i,-} = \frac{1}{2} \left[1 - \frac{\left(\frac{B_i}{2} - \Gamma_i - 1 \right)}{\sqrt{\left(\frac{B_i}{2} - \Gamma_i - 1 \right)^2 + 2B_i}} \right],$$

$$\delta_{i,+} = \frac{1}{2} \left[1 - \frac{\left(\Gamma_{i+1} - \frac{B_i}{2} + 1 \right)}{\sqrt{\left(\Gamma_{i+1} - \frac{B_i}{2} + 1 \right)^2 + 2B_i}} \right], \quad B_i = 2 \left(\frac{\Gamma_{i+1}^{\frac{\alpha}{2}} + \Gamma_i^{\frac{\alpha}{2}}}{2} \right)^{\frac{2}{\alpha}}$$

- $\Gamma_i < \frac{B_i}{2} < \Gamma_{i+1}, \quad i = 1, \dots, L-1$

- B_i increases with increase in α

- Note that

$$\lim_{\alpha \rightarrow -\infty} \frac{B_i}{2} = \Gamma_i = \min(\Gamma_{i+1}, \Gamma_i)$$

$$\left. \frac{B_i}{2} \right|_{\alpha=-2} = \left(\frac{\Gamma_{i+1}^{-1} + \Gamma_i^{-1}}{2} \right)^{-1} = \text{harmonic mean of } \Gamma_{i+1} \text{ and } \Gamma_i$$

$$\lim_{\alpha \rightarrow 0} \frac{B_i}{2} = \sqrt{\Gamma_{i+1} \Gamma_i} = \text{geometric mean of } \Gamma_{i+1} \text{ and } \Gamma_i$$

$$\left. \frac{B_i}{2} \right|_{\alpha=2} = \frac{(\Gamma_{i+1} + \Gamma_i)}{2} = \text{arithmetic mean of } \Gamma_{i+1} \text{ and } \Gamma_i$$

$$\lim_{\alpha \rightarrow \infty} \frac{B_i}{2} = \Gamma_{i+1} = \max(\Gamma_{i+1}, \Gamma_i)$$

- Under the high SNR condition

$$\left(\frac{B_i}{2} - \Gamma_i\right) \gg 1, \quad \left(\Gamma_{i+1} - \frac{B_i}{2}\right) \gg 1, \quad i = 1, \dots, L-1,$$

$$\alpha \in (-\infty, \infty) \setminus \{0\},$$

we can approximate the SEP in closed form as

$P_{\text{scnbed,e}} \approx P_{\text{scnbed,e,hiSNR}}$, where

$$P_{\text{scnbed,e,hiSNR}} = \frac{1}{L} \binom{2N-1}{N-1} \sum_{i=1}^{L-1} \left[\delta_{i,1}^N + \delta_{i,2}^N \right], \quad \text{with}$$

$$\delta_{i,1} = \frac{\left(\frac{B_i}{2}\right)}{\left(\frac{B_i}{2} - \Gamma_i\right)^2}, \quad \delta_{i,2} = \frac{\left(\frac{B_i}{2}\right)}{\left(\Gamma_{i+1} - \frac{B_i}{2}\right)^2}$$

- *Table 1: Values of $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ ($L = 4$)*

L	Γ_{av}	$[\Gamma_1, \dots, \Gamma_L]$	N	$\alpha_{\min, \text{highSNR}}$	$\alpha_{\min}$
4	200	$(800/10)[0, 1, 3, 6]$	2	1.6799	1.72
4	200	$(800/10)[0, 1, 3, 6]$	4	1.8188	1.88
4	200	$(800/10)[0, 1, 3, 6]$	8	1.9030	1.97
4	200	$(800/10)[0, 1, 3, 6]$	16	1.9497	2.02
4	200	$(800/30)[1, 4, 9, 16]$	2	1.6109	1.70
4	200	$(800/30)[1, 4, 9, 16]$	4	1.7841	1.90
4	200	$(800/30)[1, 4, 9, 16]$	8	1.8859	2.02
4	200	$(800/30)[1, 4, 9, 16]$	16	1.9414	2.09

- Tables 1, 2, and 3 show values of $\alpha_{\min, \text{highSNR}}$ (obtained from $P_{\text{scnbed}, \text{e}, \text{hiSNR}}$) and values of $\alpha_{\min}$ (obtained from the exact SEP $P_{\text{scnbed}, \text{e}}$) for different values of $\Gamma_1, \dots, \Gamma_L$ and N

- *Table 2: Values of $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ ($L = 8$)*

L	Γ_{av}	$[\Gamma_1, \dots, \Gamma_L]$	N	$\alpha_{\min, \text{highSNR}}$	$\alpha_{\min}$
8	800	$(6400/84)[0, 1, 3, 6, 10, 15, 21, 28]$	2	1.6752	1.72
8	800	$(6400/84)[0, 1, 3, 6, 10, 15, 21, 28]$	4	1.8160	1.88
8	800	$(6400/84)[0, 1, 3, 6, 10, 15, 21, 28]$	8	1.9014	1.97
8	800	$(6400/84)[0, 1, 3, 6, 10, 15, 21, 28]$	16	1.9489	2.03

- *Table 3: Values of $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ ($L = 8$)*

L	Γ_{av}	$[\Gamma_1, \dots, \Gamma_L]$	N	$\alpha_{\min, \text{highSNR}}$	$\alpha_{\min}$
8	800	(6400/204)[1, 4, 9, 16, 25, 36, 49, 64]	2	1.6095	1.69
8	800	(6400/204)[1, 4, 9, 16, 25, 36, 49, 64]	4	1.7834	1.89
8	800	(6400/204)[1, 4, 9, 16, 25, 36, 49, 64]	8	1.8856	2.00
8	800	(6400/204)[1, 4, 9, 16, 25, 36, 49, 64]	16	1.9413	2.07

- We find that
 - the values of $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ are reasonably close
 - $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ increase with increase in N
 - $\alpha_{\min, \text{highSNR}}$ and $\alpha_{\min}$ approach 2 as N becomes large
- It can also be shown that

$$\alpha_{\min, \text{highSNR}, N \gg 1} = 2$$

- For low and moderate SNR regimes and small N , the SEP minimizing values of α will deviate significantly from the value of 2

Level Optimization

- We optimize the signal amplitude levels so as to minimize the SEP P_e subject to a total energy constraint
- Optimization problem:

$$\begin{array}{ll} \min_{\sqrt{E_1}, \dots, \sqrt{E_L}} P_e & \text{OR} \quad \min_{\Gamma_1, \dots, \Gamma_L} P_e \\ \sum_{i=1}^L E_i = L E_{\text{av}} & \sum_{i=1}^L \Gamma_i = L \Gamma_{\text{av}} \\ 0 \leq E_1 < \dots < E_L & 0 \leq \Gamma_1 < \dots < \Gamma_L \end{array}$$

- $\Gamma_i = \frac{E_i \sigma_h^2}{\sigma_n^2}, \quad i = 1, \dots, L, \quad \Gamma_{\text{av}} = \frac{E_{\text{av}} \sigma_h^2}{\sigma_n^2}$

- For the *fully noncoherent optimum receiver*, the optimal ASK constellation for minimizing the SEP is not uniformly spaced but should follow an approximately geometric progression (GP) spacing at high SNRs, with first level zero
- $\Gamma_{\text{av}} \gg 1$, $\Gamma_1 = 0$, $\Gamma_i \gg 1$, $i = 2, \dots, L$
- Asymptotic optimal symbol amplitude levels:
 $\sqrt{E_{1,*}} = 0$, $\sqrt{E_{i,*}} = \sqrt{(LE_{\text{av}})^{(i-1)/(L-1)}}$, $i = 2, \dots, L$,
 large common ratio $\sqrt{(LE_{\text{av}})^{1/(L-1)}}$
- Asymptotic optimal symbol SNR levels:
 $\Gamma_{1,*} = 0$, $\Gamma_{i,*} = (L\Gamma_{\text{av}})^{(i-1)/(L-1)}$, $i = 2, \dots, L$,
 large common ratio $(L\Gamma_{\text{av}})^{1/(L-1)}$

- For the *WENV D receiver* (channel magnitude based suboptimum receiver), the asymptotic optimal symbol amplitude levels under the high SNR condition

$\Gamma_{av} \gg 1$, $\Gamma_1 = 0$, $\Gamma_i \gg 1$, $i = 2, \dots, L$,
and the large N condition $N \gg 1$ are given as

$$\sqrt{E_{1,*}} = 0, \quad \sqrt{E_{i,*}} = \frac{R_{i-2} \sqrt{L E_{av}}}{\sqrt{\left(1 + \sum_{\ell=1}^{L-2} R_{\ell}^2\right)}}, \quad i = 2, \dots, L,$$

where $R_0 = 1$ and R_i , $i \geq 1$ follows the recurrence

$$R_i = \frac{1}{2} \left(\sqrt{[16R_{i-1}^2 + 1]} - [2R_{i-1} - 1] \right),$$

$$i = 1, \dots, L - 2$$

- For the *ED receiver* (channel magnitude based suboptimum receiver), the asymptotic optimal symbol amplitude levels under the conditions

$\Gamma_{\text{av}} \gg 1$, $\Gamma_1 = 0$, $\Gamma_i \gg 1$, $i = 2, \dots, L$, $N \gg 1$
(high SNR and large N) are given as

$$\sqrt{E_{1,*}} = 0, \quad \sqrt{E_{i,*}} = \sqrt{\frac{3i(i-1)}{(L^2-1)} E_{\text{av}}}, \quad i = 2, \dots, L$$

- Note that

$$E_{i,*} - E_{i-1,*} = \frac{6(i-1)E_{\text{av}}}{(L^2-1)}, \quad i = 2, \dots, L$$

- Level optimization for the ED receiver for high SNR and large N results in *symbol energy level differences in arithmetic progression (AP)*

- For the *SCNBED* receiver (with parameter α , $\alpha \in (-\infty, \infty) \setminus \{0\}$), we consider the high SNR condition given by

$$\left(\left(\frac{\Gamma_{i+1}^{\frac{\alpha}{2}} + \Gamma_i^{\frac{\alpha}{2}}}{2} \right)^{\frac{2}{\alpha}} - \Gamma_i \right) \gg 1,$$

$$\left(\Gamma_{i+1} - \left(\frac{\Gamma_{i+1}^{\frac{\alpha}{2}} + \Gamma_i^{\frac{\alpha}{2}}}{2} \right)^{\frac{2}{\alpha}} \right) \gg 1, \quad i = 1, \dots, L-1,$$

and the large N condition $N \gg 1$

- Denote the quantities $\rho_1, \dots, \rho_{L-1}$ as

$$\rho_i = \frac{\Gamma_{i+1}}{\Gamma_i} = \frac{E_{i+1}}{E_i}, \quad i = 1, \dots, L-1$$

- When $\alpha > 2$, under the conditions of high SNR and large N , the asymptotic optimal $\rho_1, \dots, \rho_{L-1}$, which are denoted as $\rho_{1,*}, \dots, \rho_{L-1,*}$, respectively, are expressed by $1/\rho_{1,*} = 0$ and the $L - 2$ nonlinear equations (which can be solved numerically)

$$\left(\frac{\rho_{2,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}} = \frac{\left(\rho_{2,*} - \left(\frac{\rho_{2,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}}\right)^2}{\left(2^{\frac{1}{\alpha}} - 2^{-\frac{1}{\alpha}}\right)^2},$$

$$\left(\frac{\rho_{\ell+1,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}} = \frac{\rho_{\ell,*} \left(\frac{\rho_{\ell,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}} \left(\rho_{\ell+1,*} - \left(\frac{\rho_{\ell+1,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}}\right)^2}{\left(\rho_{\ell,*} - \left(\frac{\rho_{\ell,*}^{\frac{\alpha}{2}} + 1}{2}\right)^{\frac{2}{\alpha}}\right)^2},$$

$$\ell = 2, \dots, L - 2$$

- The asymptotic optimal symbol SNR levels under the conditions of high SNR and large N when $\alpha > 2$ are given by $\Gamma_{1,*} = 0$,

$$\Gamma_{2,*} = \frac{L\Gamma_{\text{av}}}{\left(1 + \sum_{j=2}^{L-1} \left[\prod_{k=2}^j \rho_{k,*} \right]\right)},$$

$$\Gamma_{i,*} = \frac{\left(\prod_{\ell=2}^{i-1} \rho_{\ell,*}\right) L\Gamma_{\text{av}}}{\left(1 + \sum_{j=2}^{L-1} \left[\prod_{k=2}^j \rho_{k,*} \right]\right)}, \quad i = 3, \dots, L$$

- When $\alpha < 2$, under the conditions of high SNR and large N , the asymptotic optimal $\rho_1, \dots, \rho_{L-1}$, which are denoted as $\rho_{1,*}, \dots, \rho_{L-1,*}$, respectively, are expressed by the $L - 1$ nonlinear equations (which can be solved numerically)

$$2\rho_{1,*}^{\frac{2}{\alpha}} + 1 = \left(\frac{\rho_{1,*}^{\frac{\alpha}{2}} + 1}{2} \right)^{\frac{2}{\alpha}},$$

$$\left(\frac{\rho_{\ell+1,*}^{\frac{\alpha}{2}} + 1}{2} \right)^{\frac{4}{\alpha}} - 2 \left[1 + \frac{\left(\left(\frac{\rho_{\ell,*}^{\frac{\alpha}{2}} + 1}{2} \right)^{\frac{2}{\alpha}} - 1 \right)^2}{2\rho_{\ell,*} \left(\frac{\rho_{\ell,*}^{\frac{\alpha}{2}} + 1}{2} \right)^{\frac{2}{\alpha}}} \right] \left(\frac{\rho_{\ell+1,*}^{\frac{\alpha}{2}} + 1}{2} \right)^{\frac{2}{\alpha}} + 1 = 0,$$

$$\ell = 1, \dots, L - 2$$

- The asymptotic optimal symbol SNR levels under the conditions of high SNR and large N when $\alpha < 2$ are given by

$$\Gamma_{1,*} = \frac{L\Gamma_{\text{av}}}{\left(1 + \sum_{j=1}^{L-1} \left[\prod_{k=1}^j \rho_{k,*} \right]\right)},$$

$$\Gamma_{i,*} = \frac{\left(\prod_{\ell=1}^{i-1} \rho_{\ell,*}\right) L\Gamma_{\text{av}}}{\left(1 + \sum_{j=1}^{L-1} \left[\prod_{k=1}^j \rho_{k,*} \right]\right)}, \quad i = 2, \dots, L$$

- *Table 4: SCNBED receiver: Values of $\rho_{1,*}$ for different values of α when $\alpha < 2$*

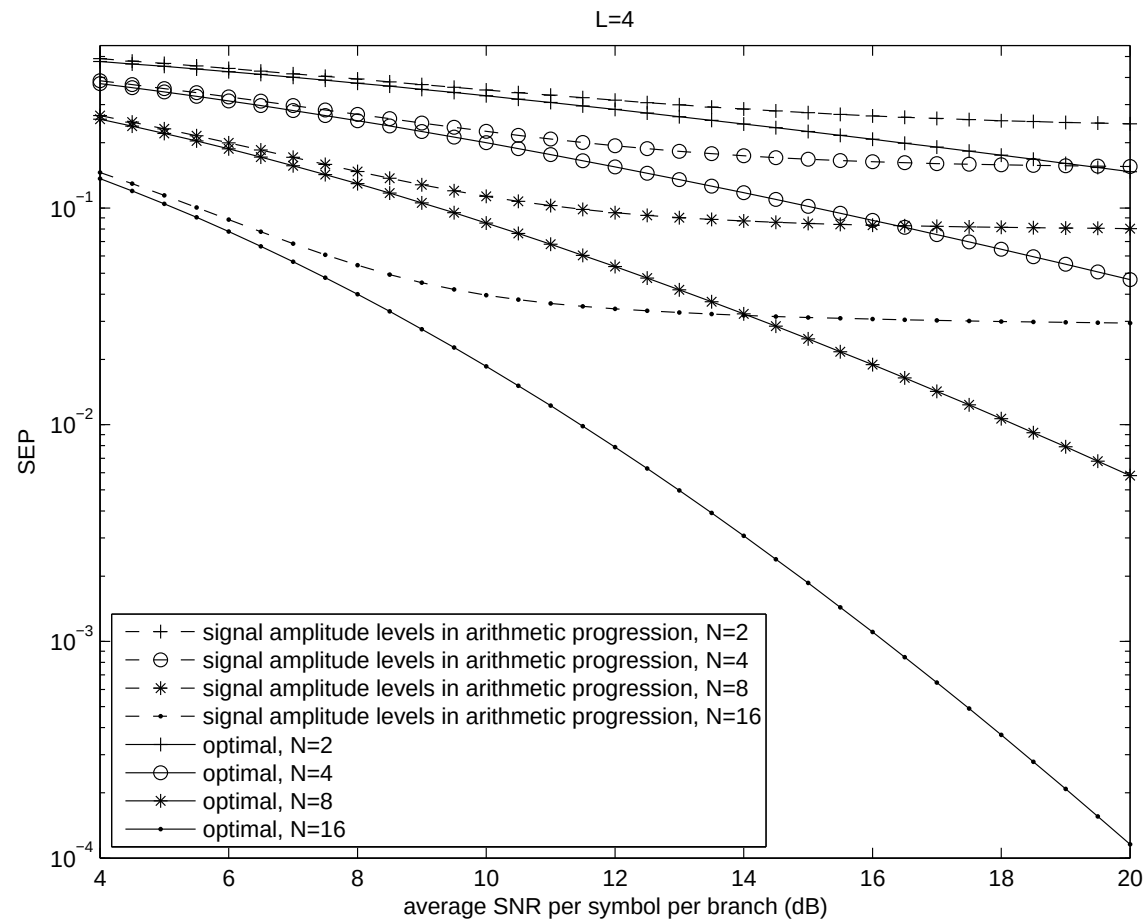
α	$\rho_{1,*}$
-4	2.5214
-2	$\frac{4}{\sqrt{17}-3} = 3.5616$
-1	4.9264
0.5	15.5792
1	$21 + 12\sqrt{3} = 41.7846$
1.2	83.4442
1.4	256.8098
1.6	2.4176×10^3
1.8	2.2650×10^6

- We find from Table 4 that $\rho_{1,*}$ increases with increase in α , and tends to infinity when $\alpha \rightarrow 2$

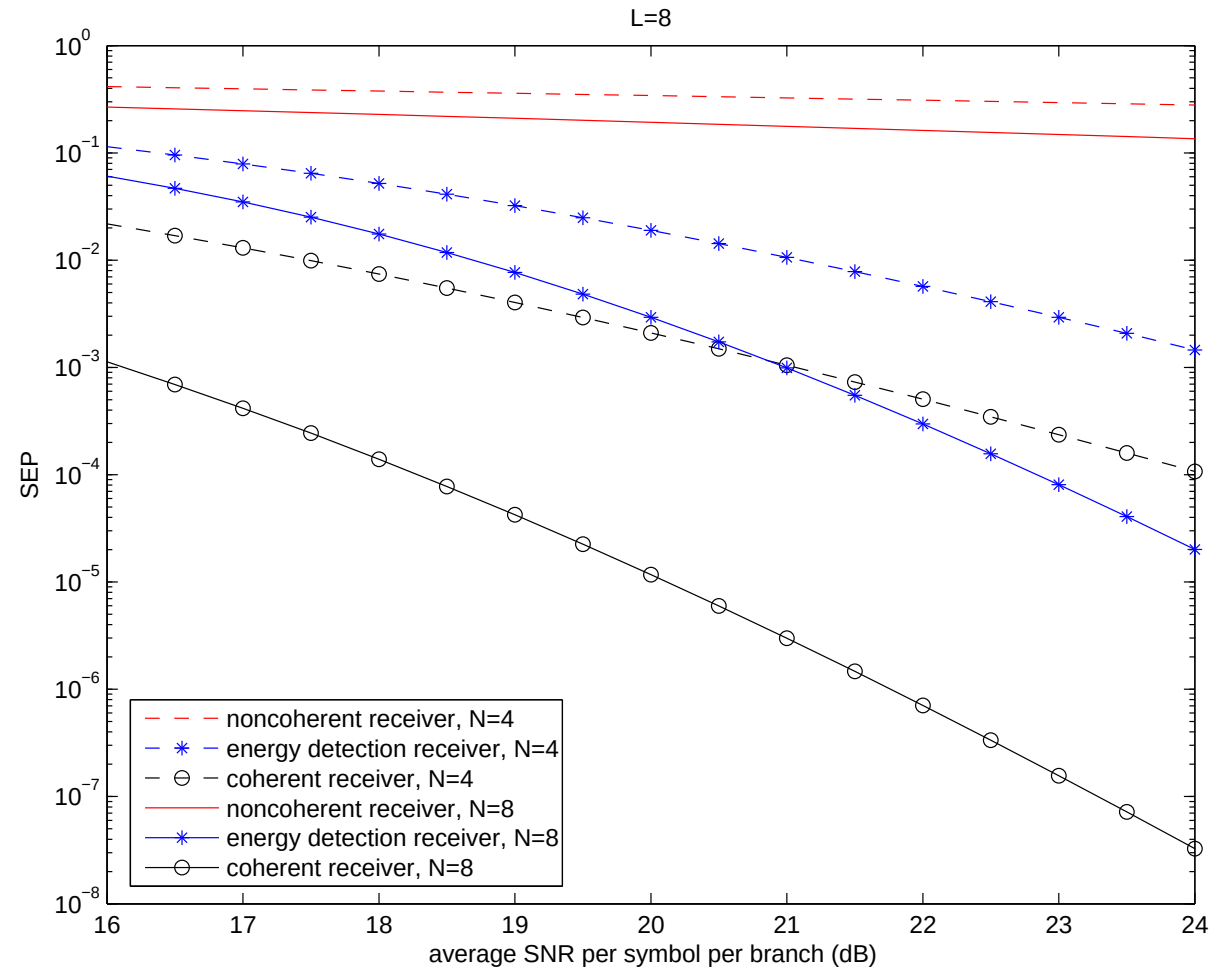
Performance

- Diversity order of fully noncoherent optimum receiver: $N/(L - 1)$
- Diversity order of WENVVD, ED, optimum CNBED, and SCNBED receivers: N
- ED performs significantly better than noncoherent approaches but it maintains the low complexity structure of noncoherent detectors
- The SEP performance of the ED receiver is much closer to that of a coherent receiver than to that of a fully noncoherent receiver
- Optimization of levels has greater impact on the WENVVD receiver than on the ED receiver

- Fig. 1: Fully noncoherent optimum receiver: Comparison of performance of 4-level ASK ($L = 4$) using optimal signal amplitude levels and signal amplitude levels in AP with varying number of diversity branches N



- Fig. 2: Performance of ED receiver (symbol amplitude levels in AP), coherent receiver (symbol amplitude levels in AP), and noncoherent receiver (symbol-plus-noise energy levels in GP)

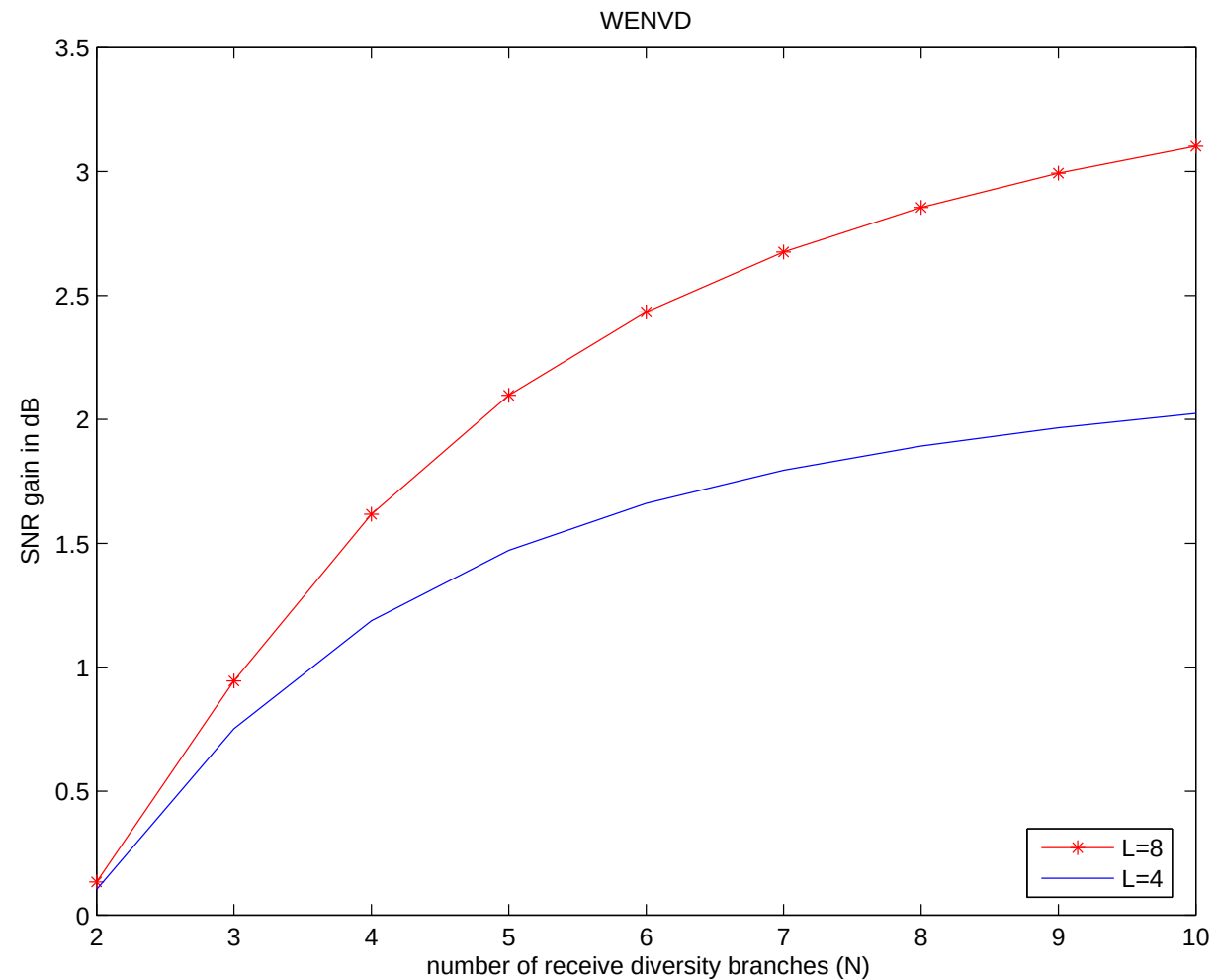


- SNR gain of WENVD receiver at high SNR obtained by using optimal levels over using amplitude levels in AP (Fig. 3):

$$\begin{aligned}
 & \text{SNR}_{\text{gain}_{\text{wenvd,highSNR}}} \\
 &= 10 \log_{10} \left(\frac{L(L-1)(2L-1)}{6 \left(1 + \sum_{\ell=1}^{L-2} R_{\ell}^2 \right)} \right) \\
 &+ \frac{10}{N} \log_{10} \left(\frac{\sum_{i=1}^{L-1} \left[\frac{1}{\left(2 - \frac{1}{(2i-1)} \right)^{2N}} + \frac{1}{\left(2 + \frac{1}{(2i-1)} \right)^{2N}} \right]}{\left[(L-1) + 9^{-N} + \sum_{i=2}^{L-1} \frac{(R_{i-1} + R_{i-2})^{2N}}{(3R_{i-1}^2 - R_{i-2}^2 - 2R_{i-1}R_{i-2})^{2N}} \right]} \right) \text{ dB}
 \end{aligned}$$

- $N \gg 1$: 2.5 dB for $L = 4$; 4 dB for $L = 8$; < 6 dB for large L

- Fig. 3: SNR gain of WENVD receiver at high SNR obtained by using optimal levels over using amplitude levels in AP

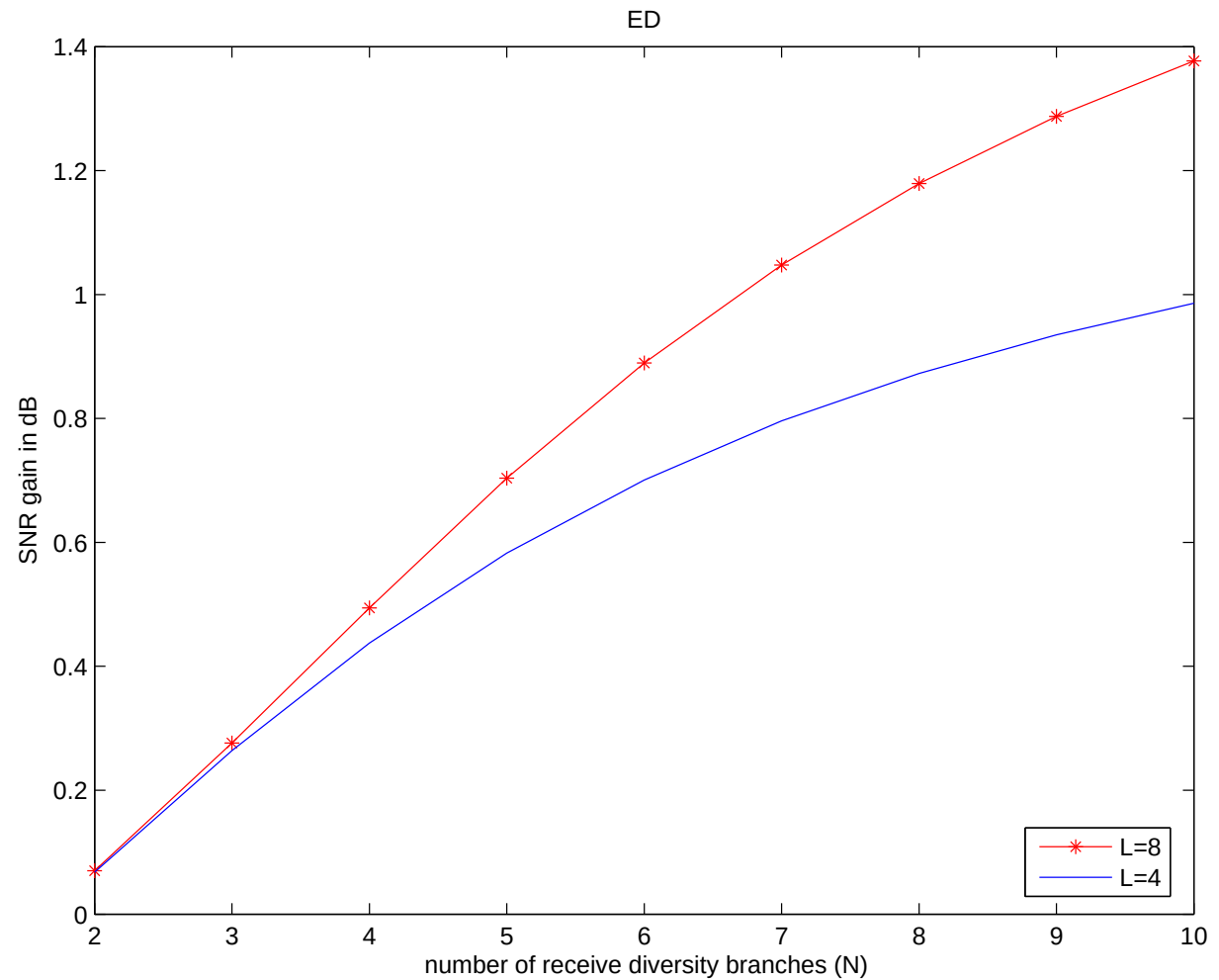


- SNR gain of ED receiver at high SNR obtained by using optimal levels over using amplitude levels in AP (Fig. 4):

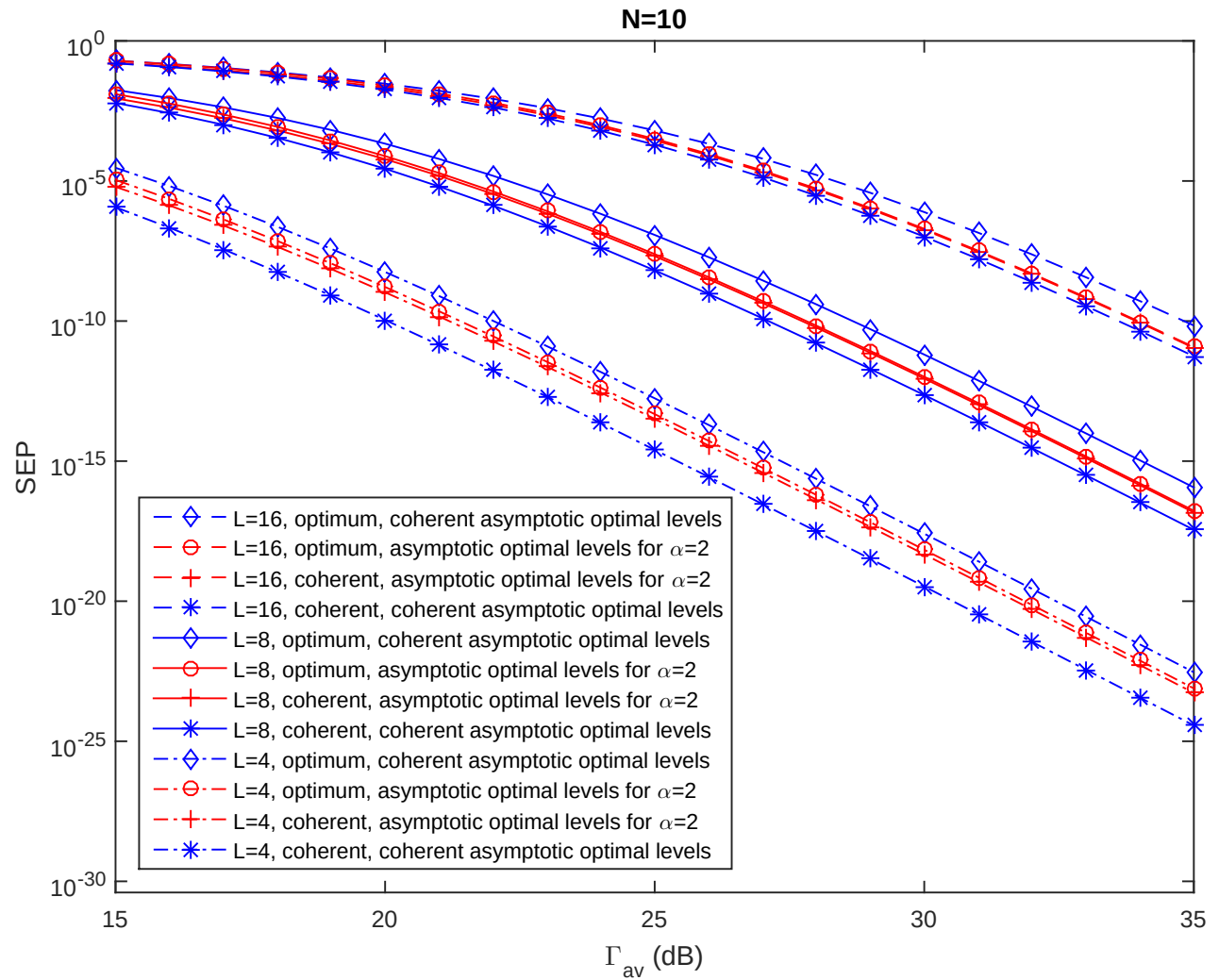
$$\begin{aligned} & \text{SNR}_{\text{gain}_{\text{ed,highSNR}}} \\ &= \frac{10}{N} \log_{10} \left(\frac{(2L-1)^N}{2^N (L+1)^N (L-1)} \sum_{i=1}^{L-1} \left[1 + \frac{1}{(2i-1)^2} \right]^N \right) \text{dB} \end{aligned}$$

- $N \gg 1$: 1.5 dB for $L = 4$; 2.2 dB for $L = 8$; 3 dB for large L

- Fig. 4: SNR gain of ED receiver at high SNR obtained by using optimal levels over using amplitude levels in AP



• Fig. 5: Error performance comparison of optimum CNBED and coherent receivers



- In Fig. 6 (performance degradation of SCNBED receiver over coherent receiver):

exact degradation in dB (per diversity branch)

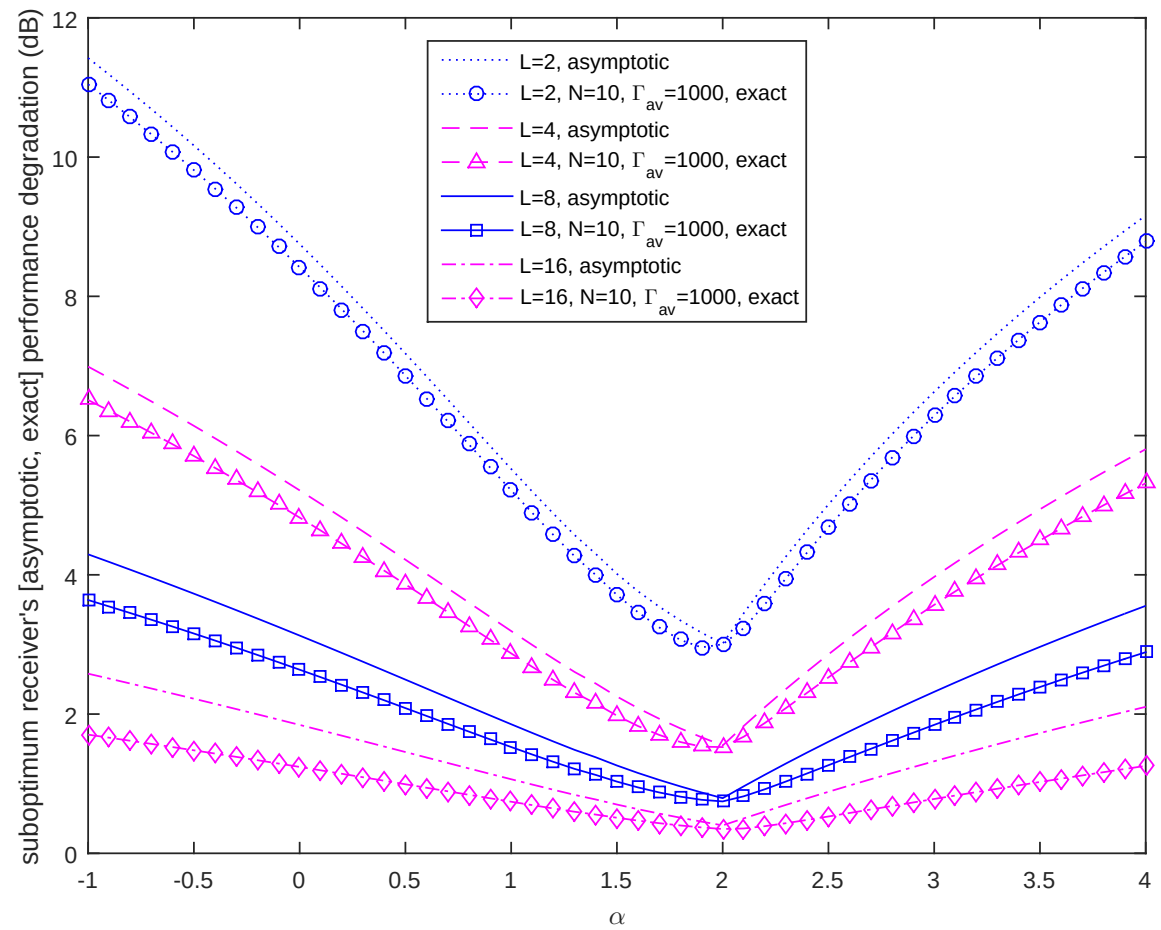
$$= \frac{10}{N} \log_{10} \left(\frac{P_{\text{scnbed},e}(\text{with asymptotic optimal levels})}{P_{\text{coh},e}(\text{with coherent asymptotic optimal levels})} \right)$$

asymptotic degradation in dB (per diversity branch)

$$= \frac{10}{N} \log_{10} \left(\frac{P_{\text{scnbed},e,\text{highSNR},N \gg 1}(\text{with asymptotic optimal levels})}{P_{\text{coh},e,\text{highSNR},N \gg 1}(\text{with coherent asymptotic optimal levels})} \right)$$

- The asymptotic performance degradation decreases with increase in L for a given α and, for a given L , it decreases with increase in α , reaches a minimum value of $10 \log_{10}((2L + 2)/(2L - 1))$ dB at $\alpha = 2$, and then increases with increase in α
- The exact performance degradation follows a similar trend

- Fig. 6: SCNBED receiver's exact and asymptotic performance degradation over coherent receiver versus α



- Advantages of using channel norm over channel element magnitudes:
 - Less computational complexity as individual elements do not need to be processed
 - Allows channel estimation process to be simplified due to only channel Euclidean norm need being estimated
 - Allows use of approximations based on central limit theorem for sum of random variables
- Optimized levels are useful for designing systems in applications such as IoT
- The CNBED approach is particularly suitable for IoT systems that employ energy harvesting where an integrated energy harvester and receiver can be developed based around an envelope detector

- For high SNR and large N , use of the SCNBED receiver with $\alpha = 2$ instead of the optimum CNBED receiver does not incur much loss in performance
- The performance degradation of CNBED receivers when compared to a coherent receiver can be made sufficiently low by appropriate choice of parameters

References

- Ranjan K. Mallik and Ross D. Murch, “Noncoherent reception of multi-level ASK in Rayleigh fading with receive diversity,” *IEEE Transactions on Communications*, vol. 62, no. 1, pp. 135-143, January 2014
- Ranjan K. Mallik, Ross D. Murch, and Yue Li, “Channel magnitude based energy detection with receive diversity for multi-level amplitude-shift keying in Rayleigh fading,” *IEEE Transactions on Communications*, vol. 65, no. 7, pp. 3079-3094, July 2017
- Ranjan K. Mallik and Ross D. Murch, “Optimal ASK levels for channel magnitude based diversity reception in Rayleigh fading,” *IEEE Transactions on Communications*, vol. 66, no. 9, pp. 4345-4360, September 2018
- Ranjan K. Mallik and Ross Murch, “Channel norm-based energy detection with diversity reception for amplitude-shift keying in Rayleigh fading,” *IEEE Transactions on Wireless Communications*, vol. 22, no. 9, pp. 5855-5870, September 2023

THANK YOU